

LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data

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Batch: SEP BATCH

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1. Objective

The objective of this project is to predict whether a patient has diabetes using Logistic Regression. The target column is Outcome:

- Outcome = 0 : No diabetes
- Outcome = 1 : Diabetes

2. Dataset

Pima Indians Diabetes Database (Kaggle dataset). File: diabetes.csv. The data has 768 patients (rows) and 9 columns: 8 input features and the Outcome column. Out of 768 patients, 500 (65.1%) have No Diabetes and 268 (34.9%) have Diabetes.

Source: <https://www.kaggle.com/datasets/hasibur013/diabetes-dataset>

3. Model

- Median imputation
- StandardScaler
- Logistic Regression (C = 0.01, max_iter = 1000, random_state = 42)

4. Dataset and Workflow

Input features:

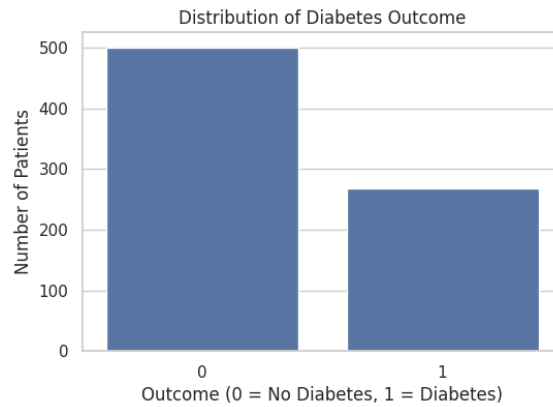
Feature	Feature
Pregnancies	BMI
Glucose	DiabetesPedigreeFunction
BloodPressure	Age
SkinThickness	Insulin

Workflow: load Kaggle CSV → inspect data → replace invalid zeros → train-test split → median imputation → scaling → train Logistic Regression → evaluate → interpret coefficients

Observation: There were no NaN values and no duplicate rows, but Glucose, BloodPressure, SkinThickness, Insulin and BMI had zeros that are not valid measurements (Insulin 374, SkinThickness 227, BloodPressure 35, BMI 11, Glucose 5). These zeros were replaced with missing values and filled with the median. The data was split 80/20 with stratify (614 training and 154 test rows), and the median and scaler were fitted on the training data only.

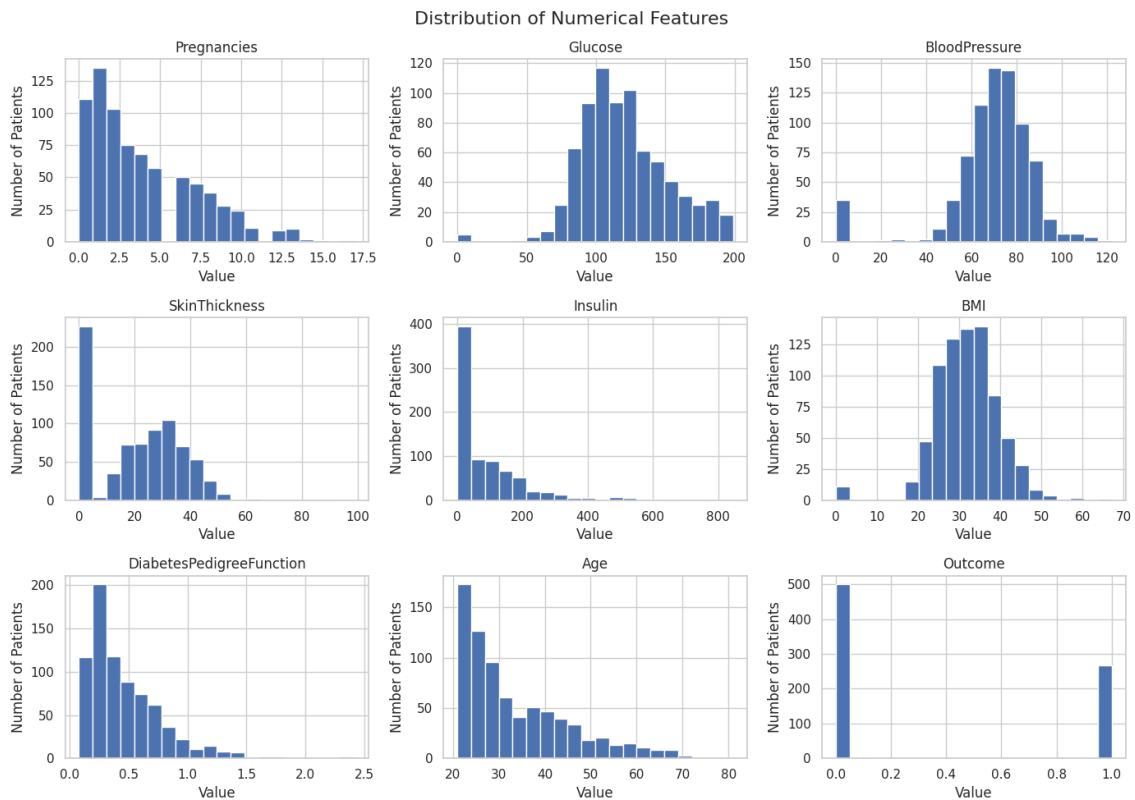
Graphs from the notebook (EDA on the original data, before the zeros were treated):

Figure 1. Distribution of Diabetes Outcome



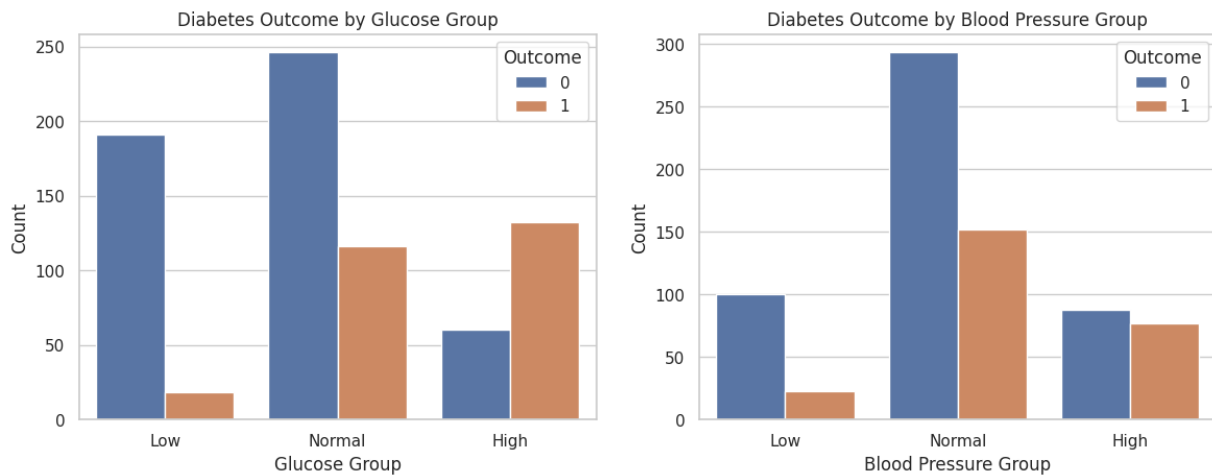
Observation: There are 500 patients with No Diabetes and 268 with Diabetes, so the data is imbalanced.

Figure 2. Distribution of Numerical Features



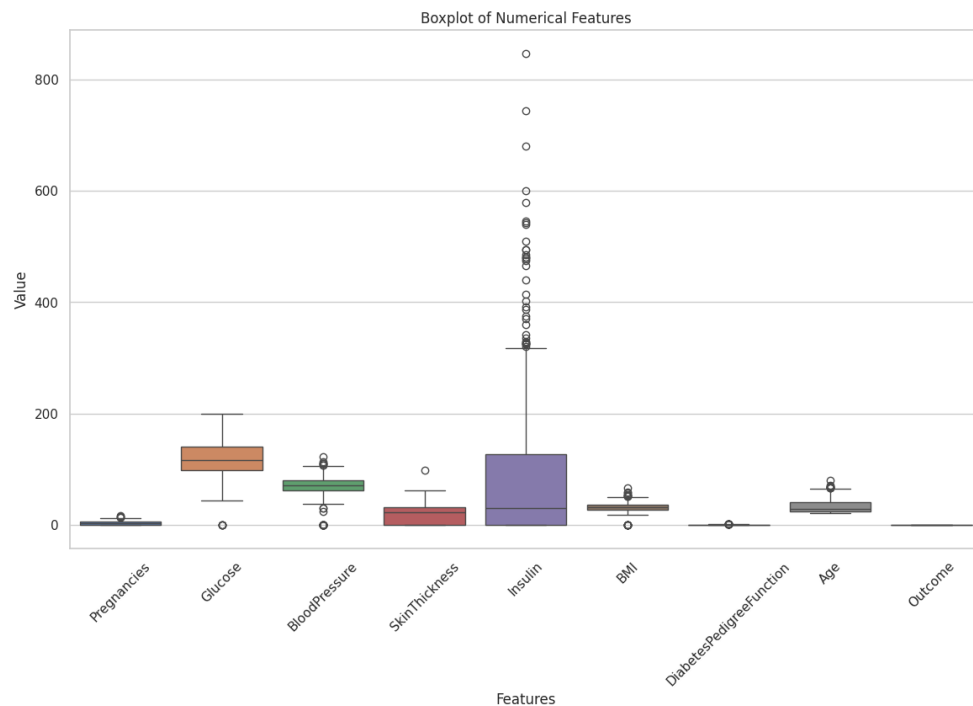
Observation: Glucose, BloodPressure and BMI look roughly bell shaped but have a small bar at 0. SkinThickness and Insulin have a very big bar at 0. Insulin has a long tail to the right. Pregnancies, DiabetesPedigreeFunction and Age are skewed to the right.

Figure 3. Diabetes Outcome by Glucose Group and Blood Pressure Group



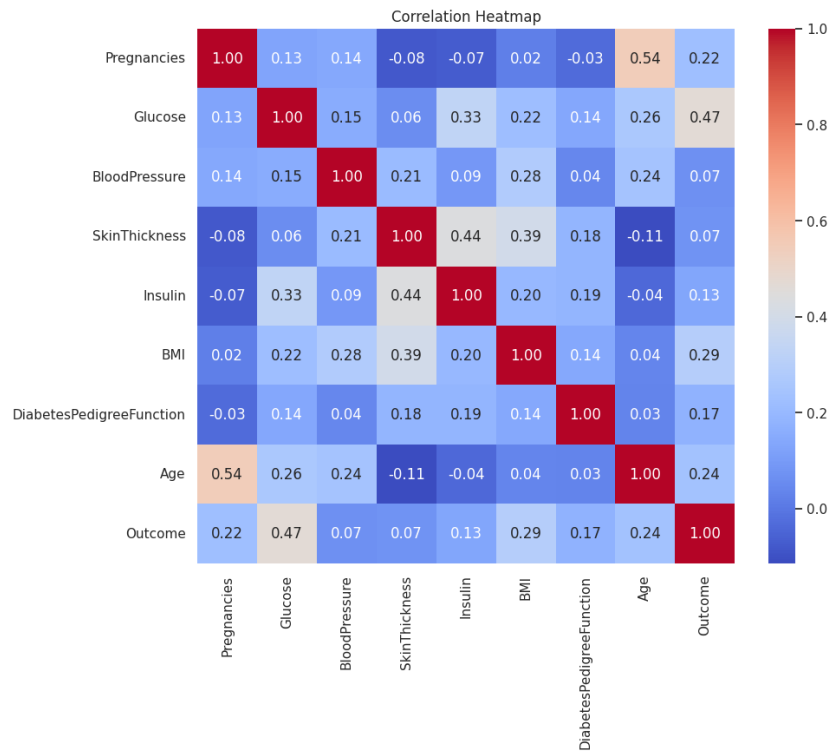
Observation: The share of diabetes patients goes up clearly with the glucose group (about 8.6% in Low, 32.0% in Normal, 68.8% in High). It also goes up with the blood pressure group (about 18.7%, 34.2%, 46.7%), but the change is smaller.

Figure 4. Boxplot of Numerical Features



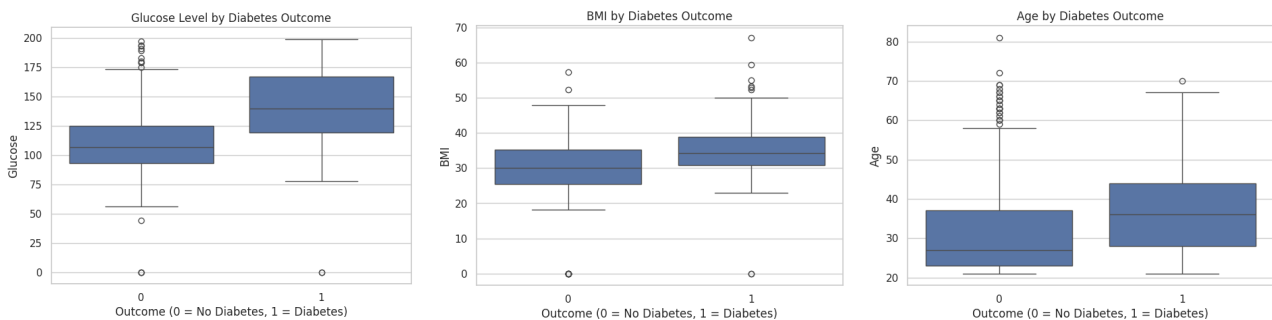
Observation: Insulin has the biggest range (up to 846) and many outliers. Glucose, BloodPressure and BMI show dots at 0, which are the invalid zeros.

Figure 5. Correlation Heatmap



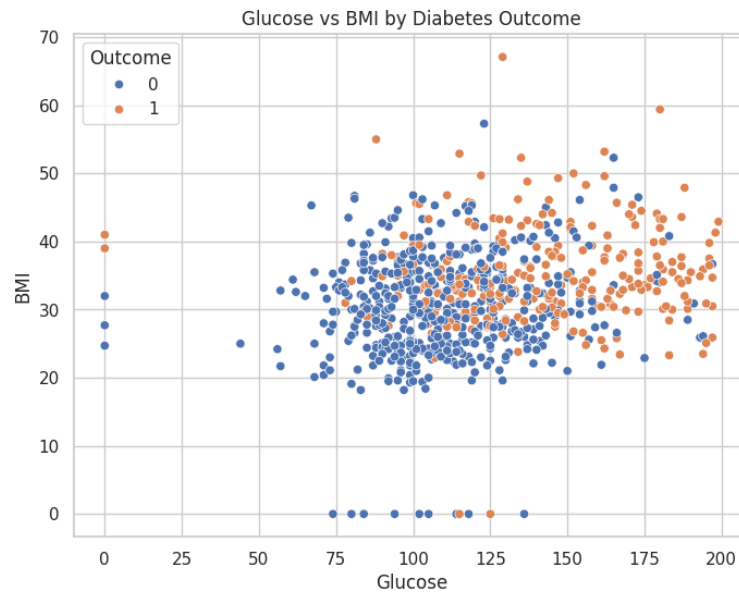
Observation: Correlation with Outcome: Glucose 0.47 (highest), BMI 0.29, Age 0.24, Pregnancies 0.22, DiabetesPedigreeFunction 0.17, Insulin 0.13, BloodPressure 0.07 and SkinThickness 0.07. No pair of input features is very highly correlated.

Figure 6. Glucose, BMI and Age by Diabetes Outcome



Observation: Median Glucose is 107 for No Diabetes and 140 for Diabetes. Median BMI is 30.05 and 34.25. Median Age is 27 and 36. Patients with diabetes are generally higher in all three.

Figure 7. Glucose vs BMI by Diabetes Outcome



Observation: Diabetes points are mostly at higher glucose values. BMI alone does not separate the two groups as clearly as glucose does.

5. Evaluation Metrics

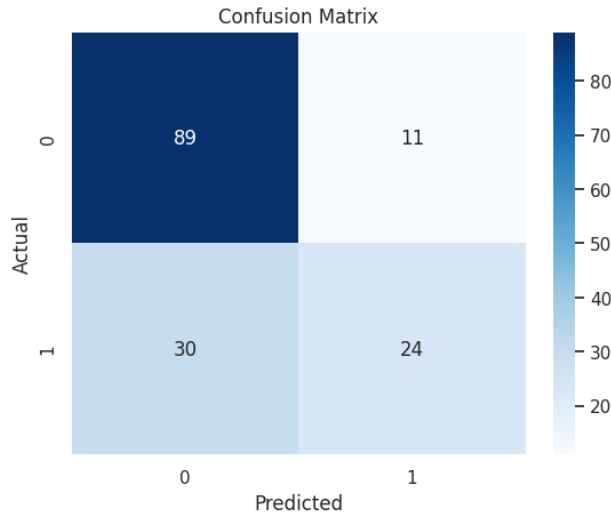
Metric	Value
Accuracy	0.73
Precision	0.69
Recall	0.44
F1-score	0.54
ROC-AUC	0.812

Observation: Precision, recall and F1-score are for the Diabetes class (Outcome = 1) on the 154 test patients. Accuracy alone can mislead here, because always predicting No Diabetes would already give about 65%.

6. Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 89	FP = 11
Actual 1	FN = 30	TP = 24

Figure 8. Confusion Matrix



Observation: $89 + 24 = 113$ test patients were predicted correctly and $11 + 30 = 41$ were wrong. The model missed 30 of the 54 patients who have diabetes.

7. Key Findings

The Logistic Regression model gave an accuracy of 0.73 on the test set, with 113 of 154 patients predicted correctly. It is better at finding patients without diabetes (89 of 100 correct) than patients with diabetes (24 of 54 correct), so the recall for diabetes is only 0.44 and 30 diabetes patients were missed. When the model does predict diabetes it is right about 69% of the time. The ROC-AUC is 0.812, which means the model ranks diabetes and non-diabetes patients reasonably well even though the accuracy is modest. In the coefficients, Glucose is the strongest feature (0.53) and BMI is next (0.30), followed by Pregnancies (0.17) and Age (0.15). This matches the EDA, where patients with diabetes had a higher median Glucose (140 vs 107) and a higher median BMI (34.25 vs 30.05). BloodPressure had the smallest coefficient (0.07).

Figure 9. Logistic Regression Feature Coefficients

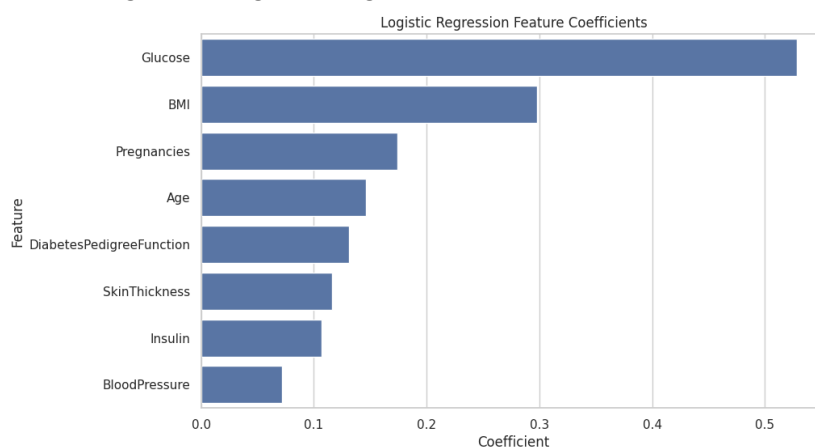
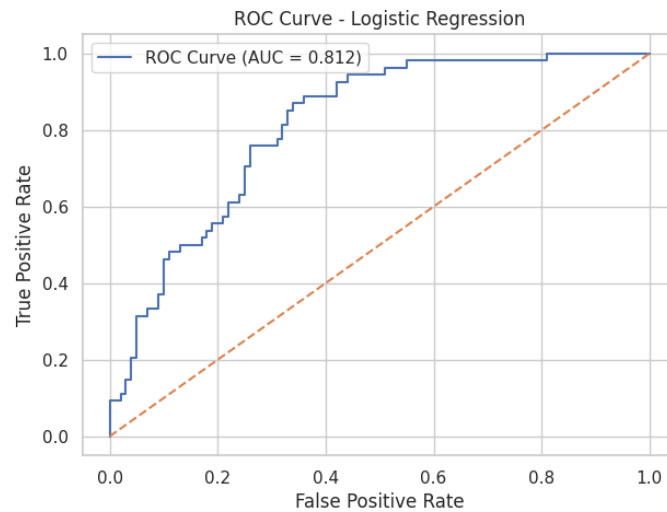


Figure 10. ROC Curve (AUC = 0.812)



8. Limitation

This is a small historical dataset (768 patients) that represents a specific population, so the results may not hold for other groups of people. The test set has only 154 patients and one train-test split was used, so the numbers can change a little with a different split. Many values were stored as zero and were filled with the median, which is only an estimate. This model is suitable for learning classification concepts, not for medical diagnosis.